scqfig gallery

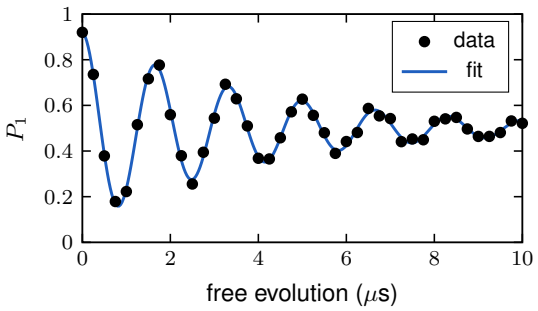
Every example in both styles: **field** (left) and **folio** (right). Build with `python examples/build_gallery.py`; each panel comes from `examples/NN_name.py`.

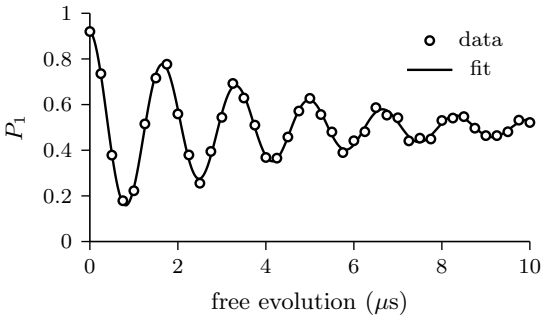
The LaTeX package

`\usepackage[field]{scqfig}` or `[folio]` defines the palette, both styles' roles (`scqFieldAccent`, `scqFolioAccent`) and the active style's roles under plain names (`scqAccent`). `\scqgraphics{t1_decay}` loads the figure of the active style, and the pgfplots styles `scqfig`, `scqfig data`, `scqfig model` and `scqfig ref` match the matplotlib output.

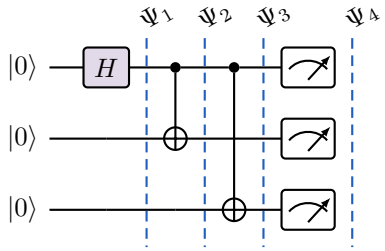
colour	field	folio
scqInk		
scqData		
scqModel		
scqSim		
scqReference		
scqAccent		
scqSignal		
scqBus		
scqWash		
scqOverlay		
scqQubitA		
scqQubitB		
scqQubitC		

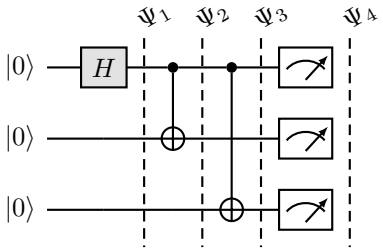
colour	field	folio
scqQubitD		
scqLevel0		
scqLevel1		
scqLevel2		
scqLevel3		
scqTileNonlinear		
scqTileCapacitive		
scqTileInductive		
scqChipPad		
scqChipResonator		
scqChipDrive		
scqChipFlux		





pgfplots axes drawn natively by LaTeX with `[scqfig field]` (left) and `[scqfig folio]` (right).





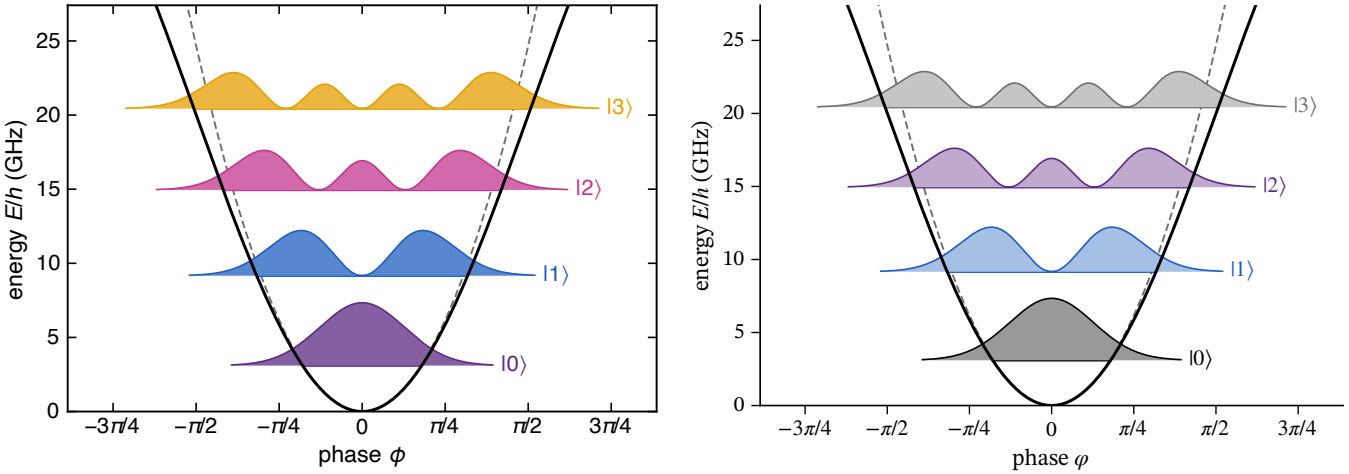
The GHZ circuit of example 78 typeset by quantikz with `[scqfig circuit field]` (left) and `[scqfig circuit folio]` (right). Each slice tilts its label with `\slice[label style=scq slice label]`.

Spectrum

01 Potential well and wavefunctions

`01_potential_well.py`

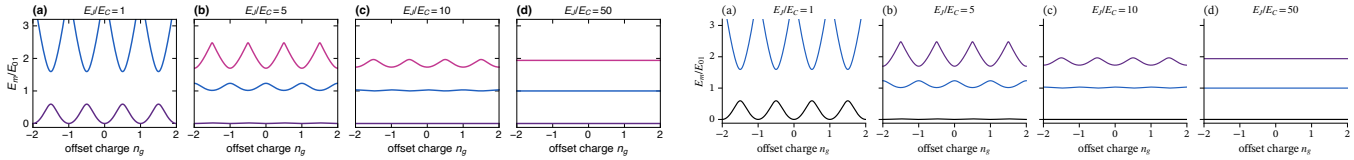
Transmon potential with the four lowest levels and their computed $|\psi(\varphi)|^2$ (charge basis, E_J/E_C = 80).



02 Charge dispersion

`02_charge_dispersion.py`

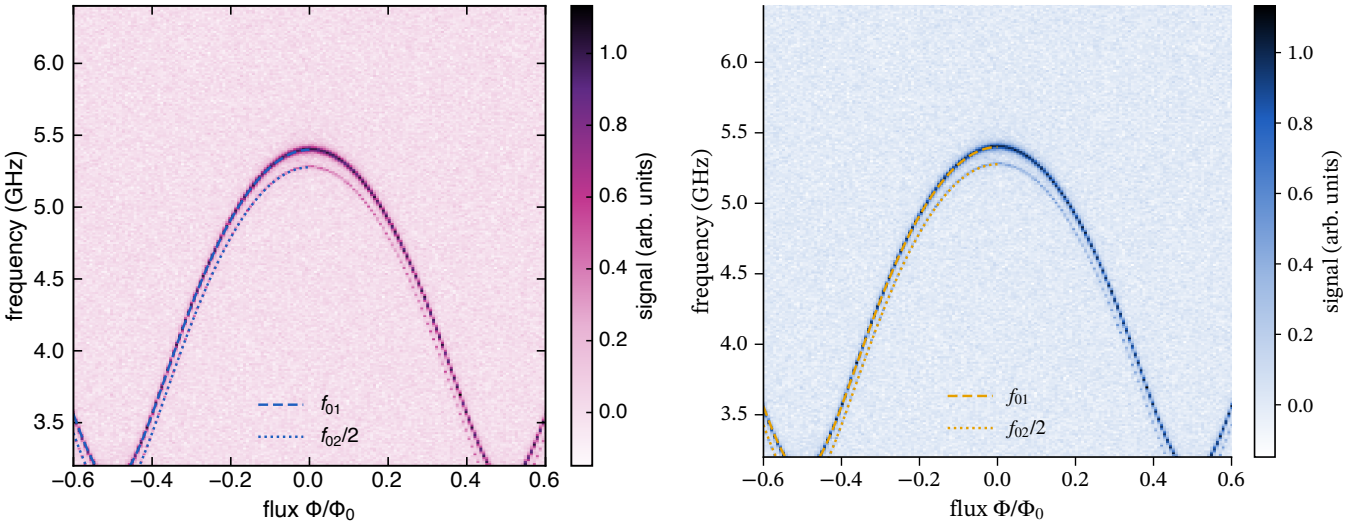
Charge dispersion of the three lowest levels as E_J/E_C grows from the Cooper-pair box to the transmon.



03 Flux-tuned spectroscopy

`03_flux_spectroscopy.py`

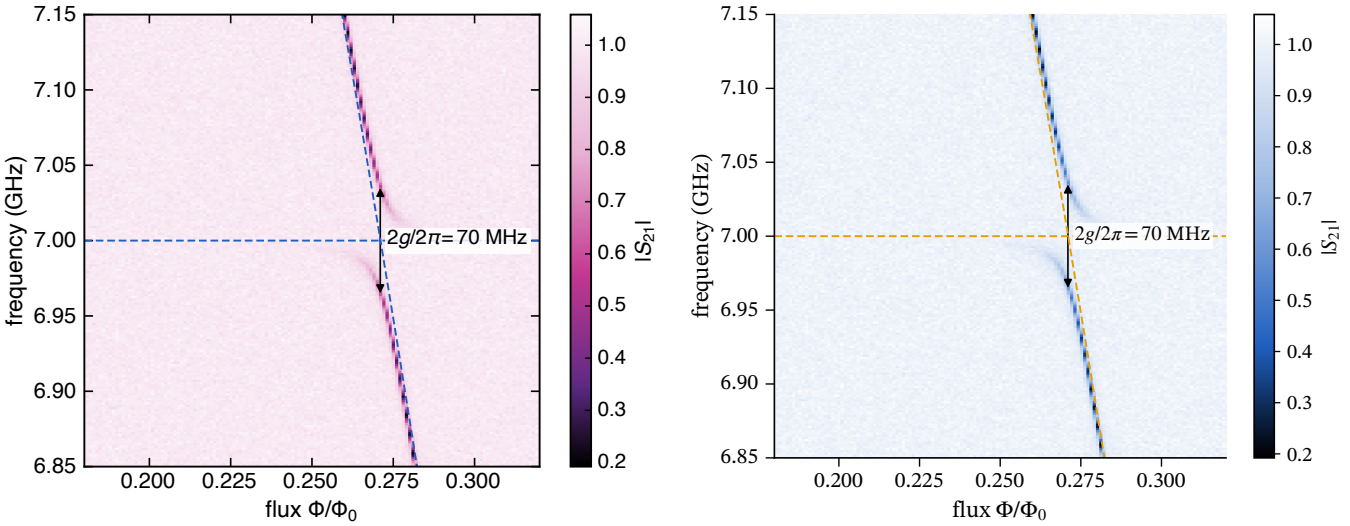
Two-tone spectroscopy of a SQUID transmon against flux, with the exact f01 and f02/2 drawn over the data (synthetic data).



04 Avoided crossing

`04_avoided_crossing.py`

Vacuum-Rabi splitting: resonator transmission as a tunable qubit crosses it, with the bare modes dashed and 2g marked (synthetic data).

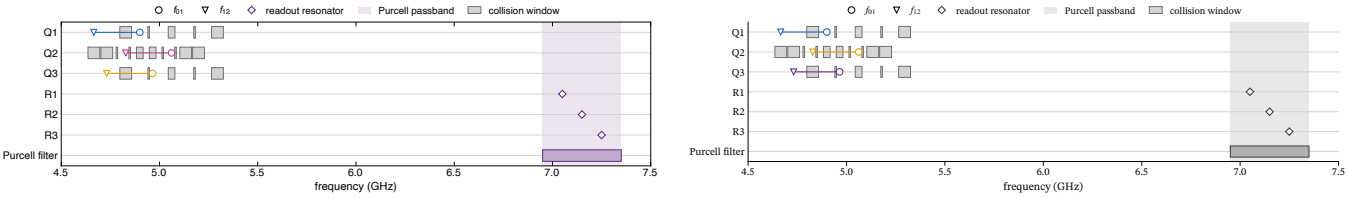


Design

10 Frequency plan

`10_frequency_plan.py`

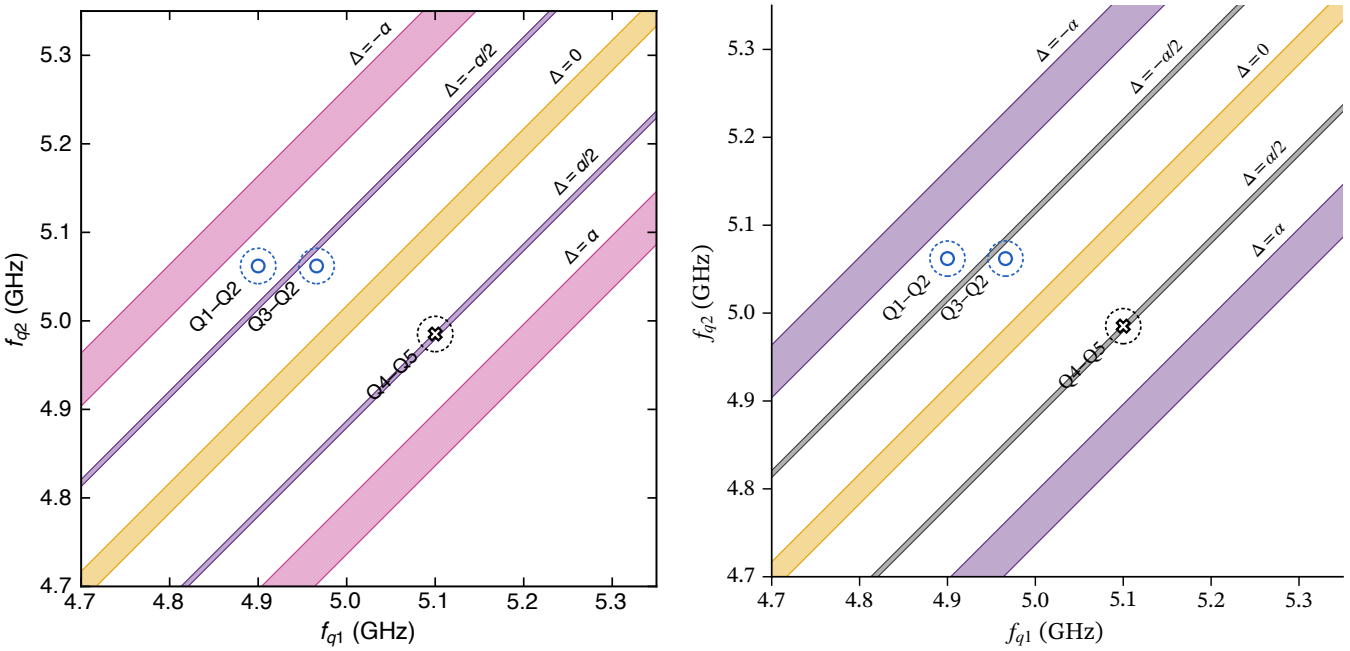
Frequency plan of a three-qubit chain: each transmon's f01 and f12, the readout resonators inside the Purcell-filter passband, and the collision windows each qubit's f01 must avoid (designed values).



11 Collision map

`11_collision_map.py`

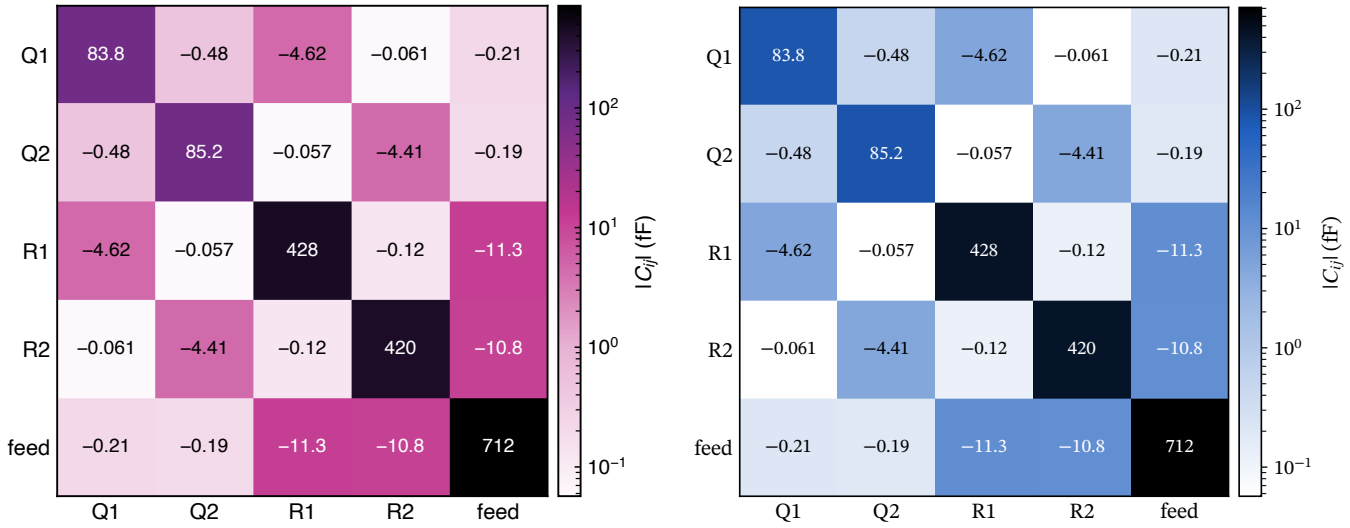
Collision map of two coupled transmons: forbidden bands of the f_q2 against f_q1 plane, with three design points and their 20 MHz fabrication scatter; one point sits on a two-photon band (designed values).



12 Capacitance matrix

`12_capacitance_matrix.py`

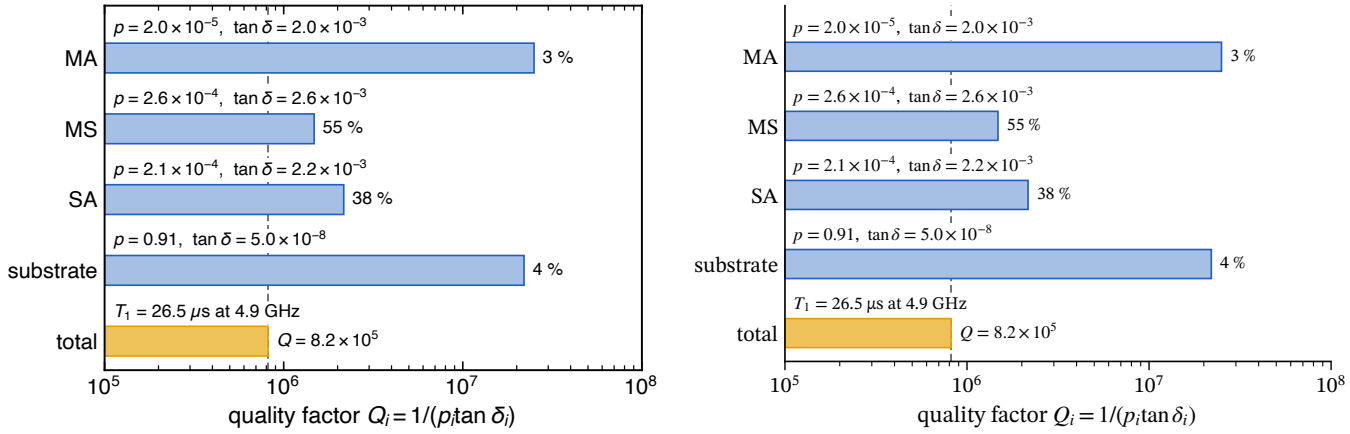
Maxwell capacitance matrix of two transmons, their readout resonators and the feedline, coloured by |C| on a log scale (simulated values).



13 Participation budget

`13_participation.py`

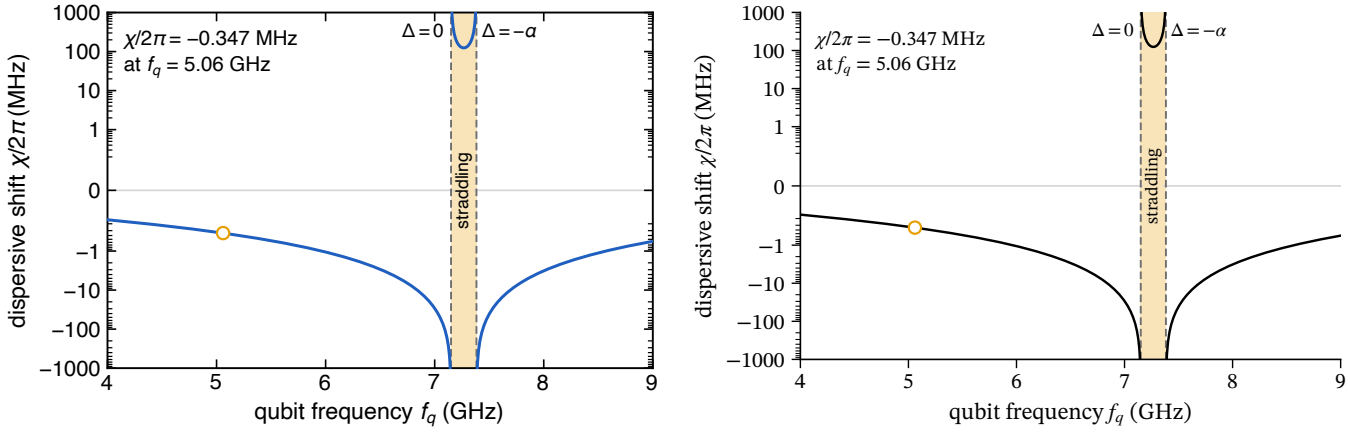
Surface-participation loss budget of a transmon: participation and loss tangent of each region, the quality factor each allows, and the total (simulated participation, assumed loss tangents).



14 Dispersive shift

`14_dispersive_shift.py`

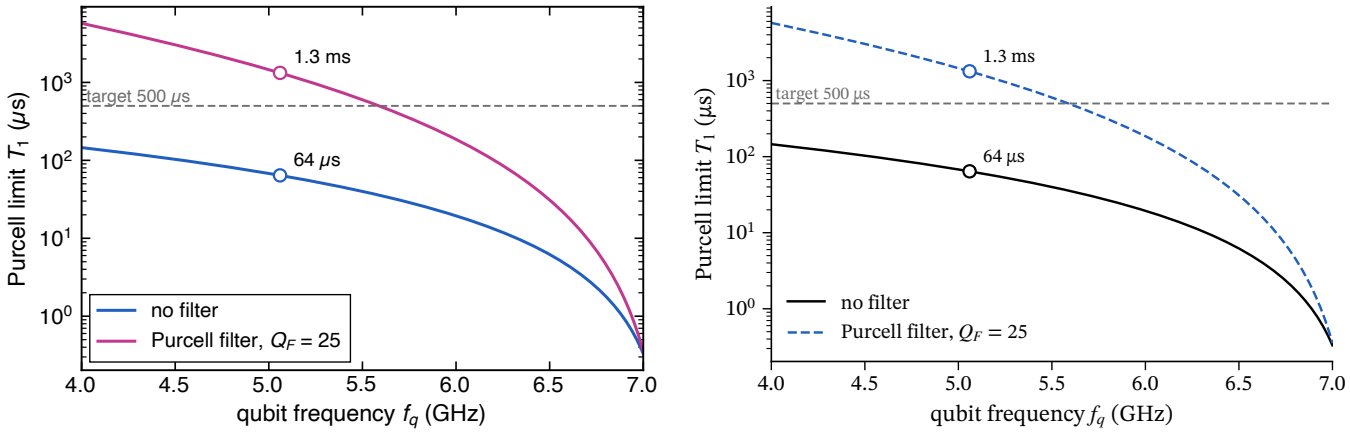
Dispersive shift of a transmon against its frequency for a 7.15 GHz resonator, g = 85 MHz and alpha = -233 MHz, with the straddling regime shaded and the design point marked (computed).



15 Purcell limit

15_purcell_limit.py

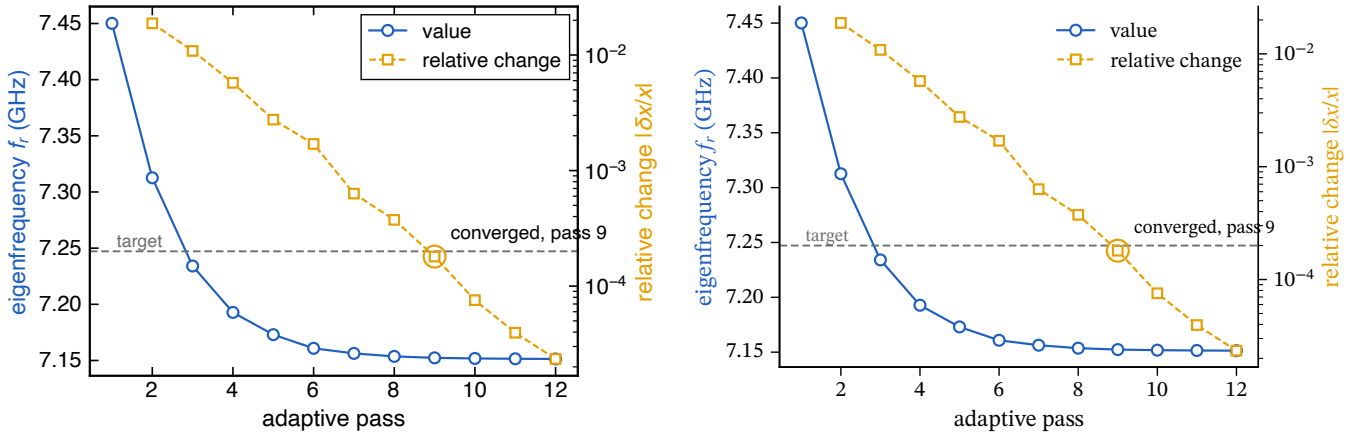
Purcell limit on T1 against qubit frequency for a 7.15 GHz readout resonator (kappa = 1.5 MHz, g = 85 MHz), without and with a Q = 25 bandpass filter (computed).



16 Convergence

16_convergence.py

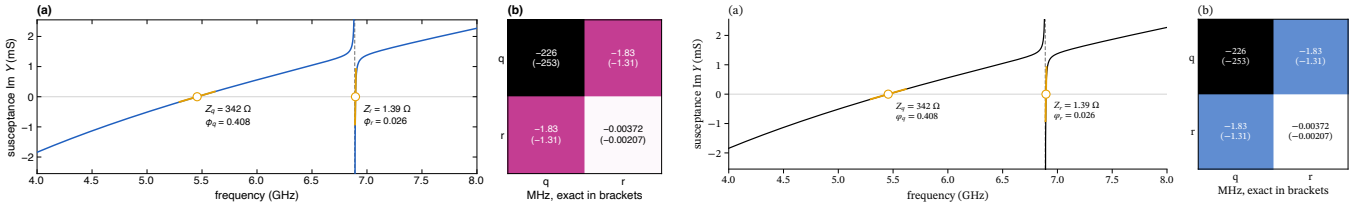
Eigenmode frequency of a readout resonator against adaptive mesh pass, with the relative change per pass and a 0.02 % target (synthetic data).



17 Black-box quantization

17_black_box.py

Black-box quantization of the transmon and readout resonator of example 72: (a) the susceptance the junction sees, whose rising zeros are the modes and whose slopes set their impedances; (b) the first-order Kerr matrix, with exact diagonalization of the same circuit in brackets (computed).

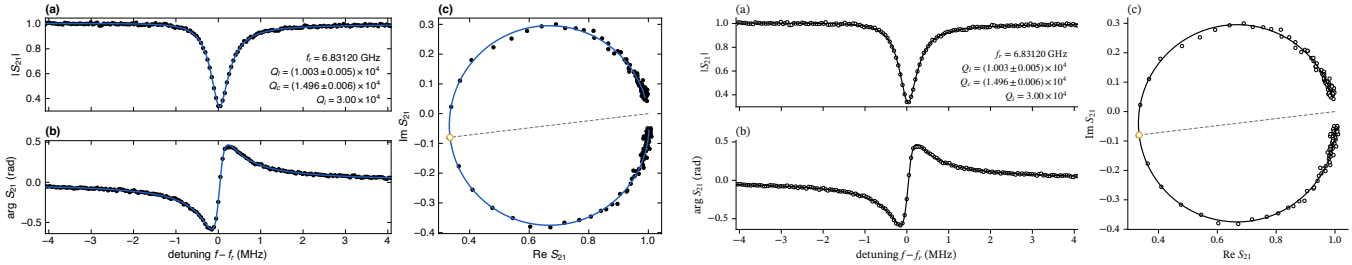


Readout

20 S21 of a resonator

20_s21_resonator.py

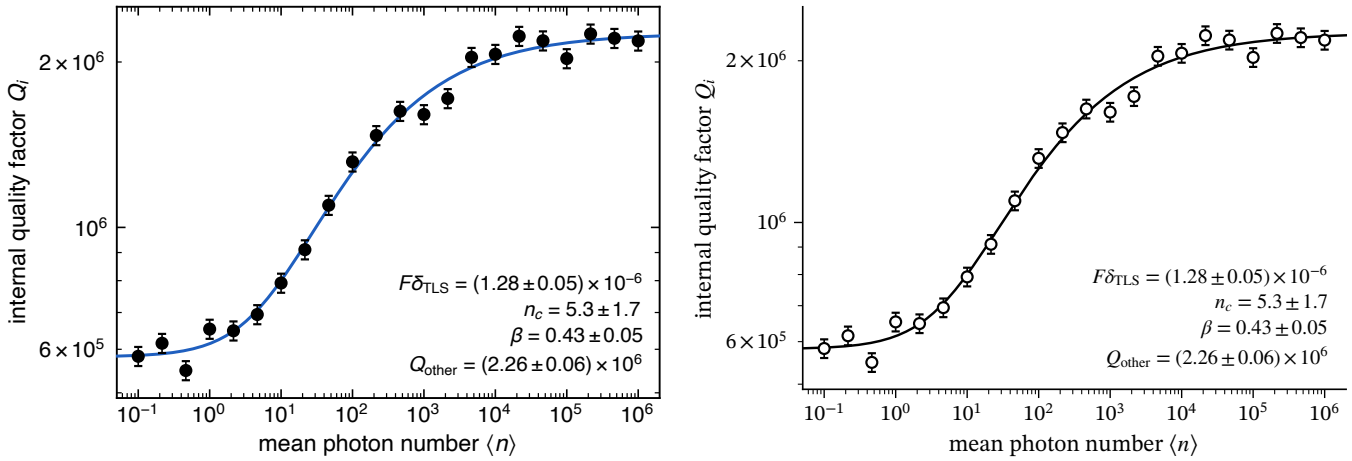
Notch-resonator transmission as magnitude, phase and circle, with one complex fit and the cable delay, amplitude and phase calibrated out (synthetic data).



21 Quality factor vs power

`21_quality_vs_power.py`

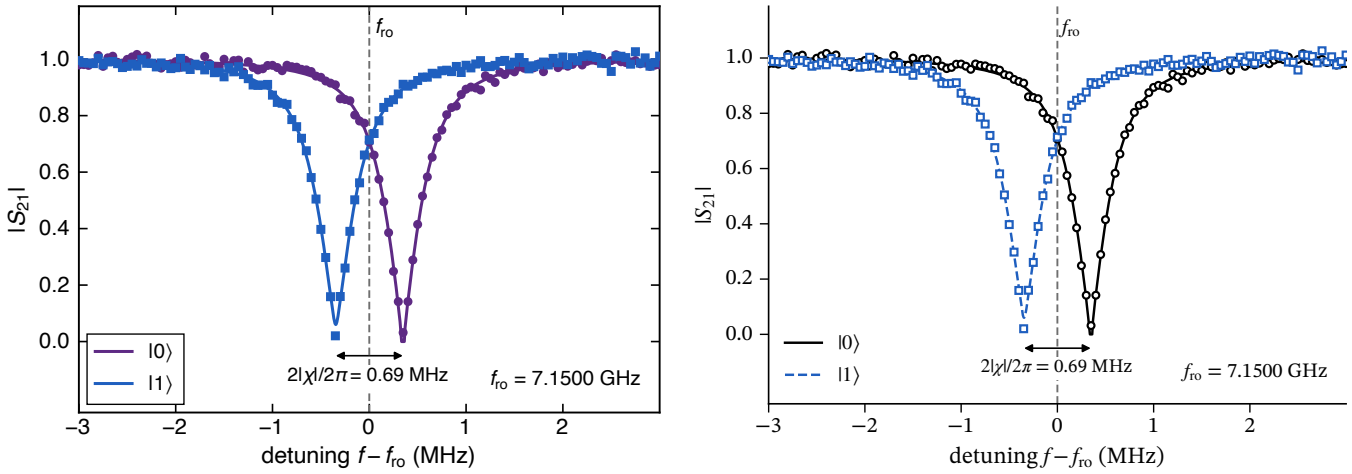
Internal quality factor of a resonator against mean photon number, with a fit of the saturable two-level-system model (synthetic data).



22 Dispersive readout

`22_dispersive_readout.py`

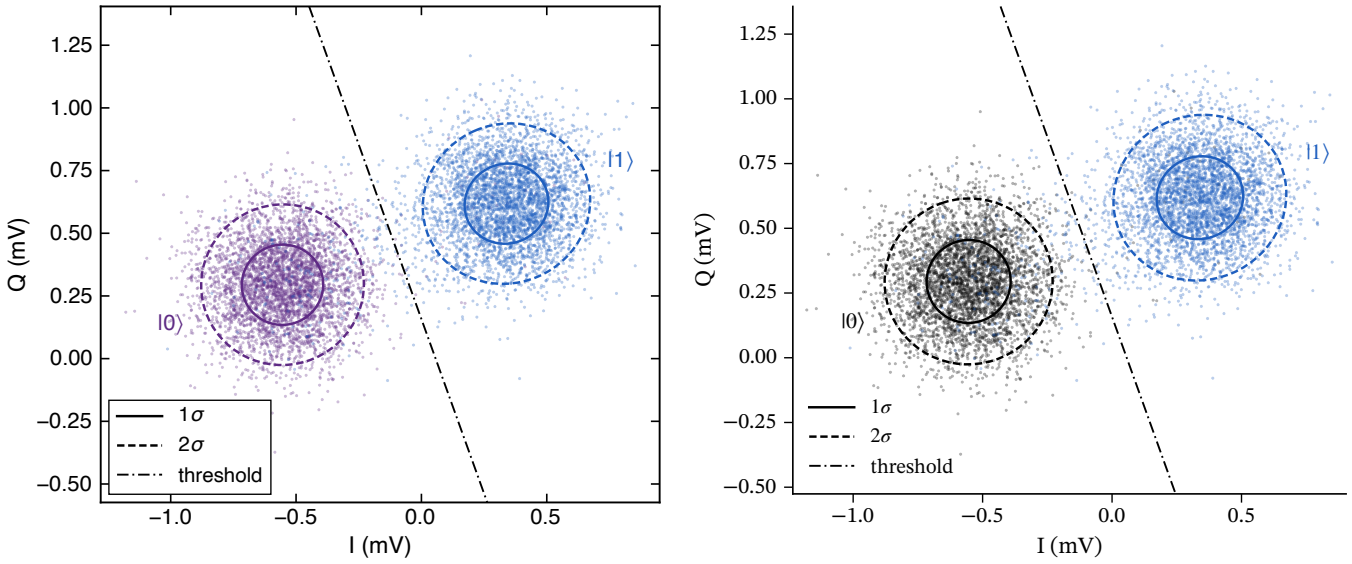
Dispersive readout: resonator transmission with the qubit in |0> and in |1>, the 2chi shift computed for g = 85 MHz and Delta = -2.09 GHz, and the readout frequency between the two dips (synthetic data).



23 IQ blobs

`23_iq_blobs.py`

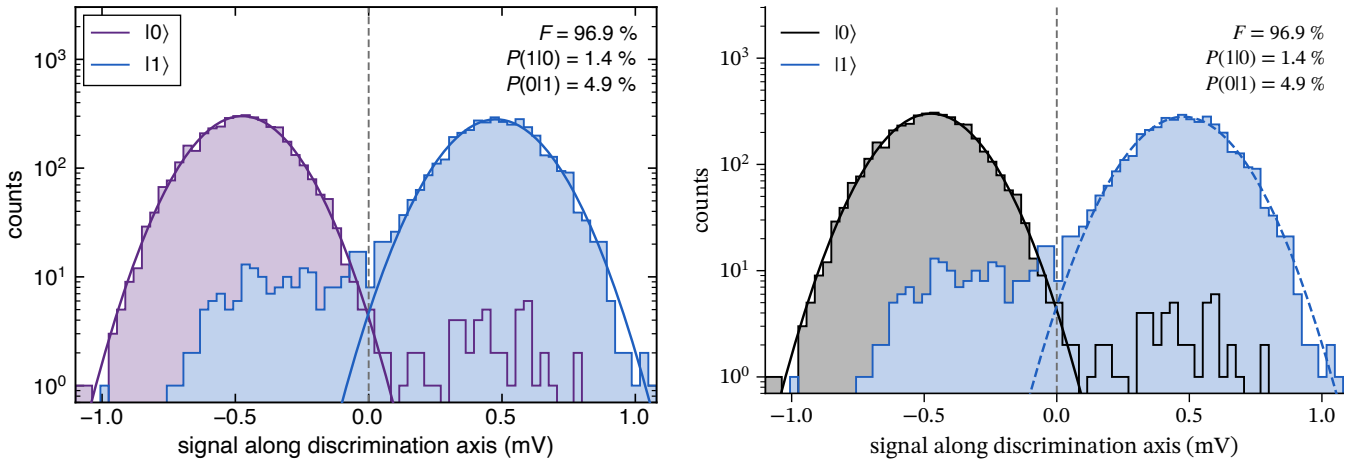
Single-shot readout in the IQ plane: 4000 shots per prepared state with 1 and 2 sigma contours and the discrimination line; the |1> shots include T1 decay during the 1.5 us integration (synthetic data).



24 Readout histogram

`24_readout_histogram.py`

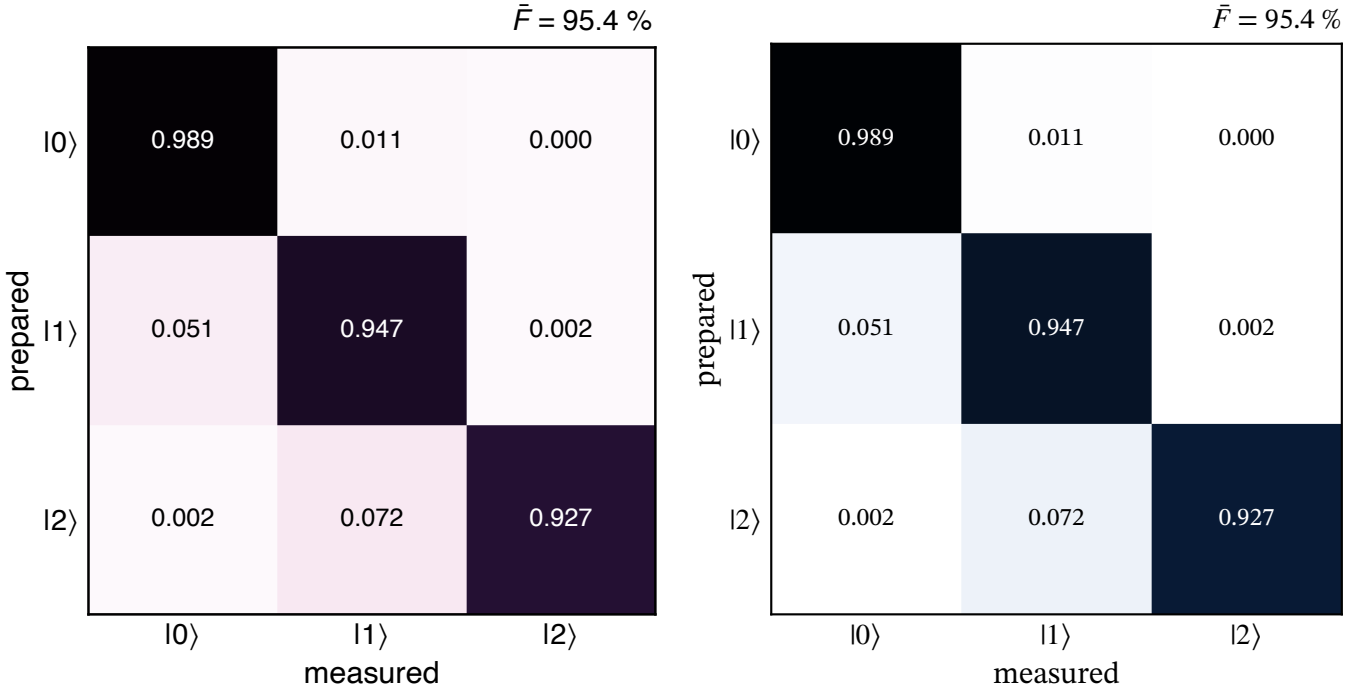
Single-shot histograms projected on the discrimination axis, with Gaussian fits to each peak and the assignment fidelity; the tail between the peaks is T1 decay during readout (synthetic data).



25 Assignment matrix

`25_assignment_matrix.py`

Three-state assignment matrix from 5000 simulated shots per prepared state, assigned to the nearest IQ centre; decay during readout fills the lower triangle (synthetic data).

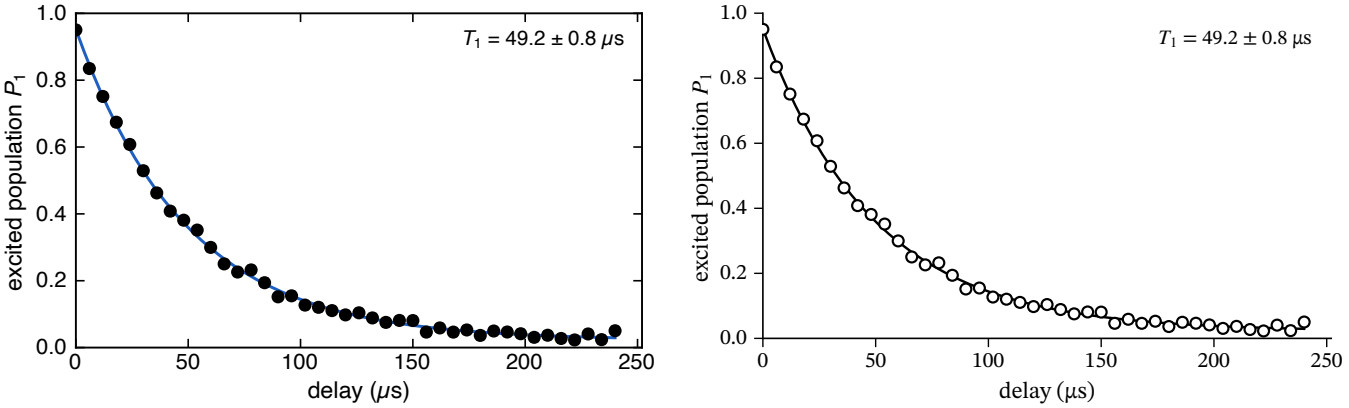


Time domain

30 T1 decay

`30_t1_decay.py`

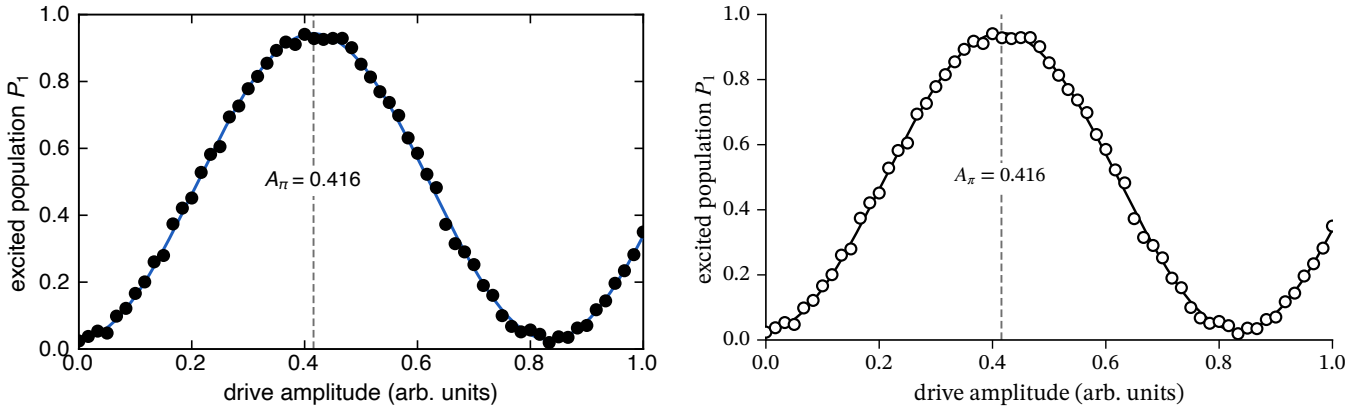
Energy relaxation: excited-state population against delay with an exponential fit (synthetic data, T1 = 48 us).



31 Power Rabi

`31_rabi.py`

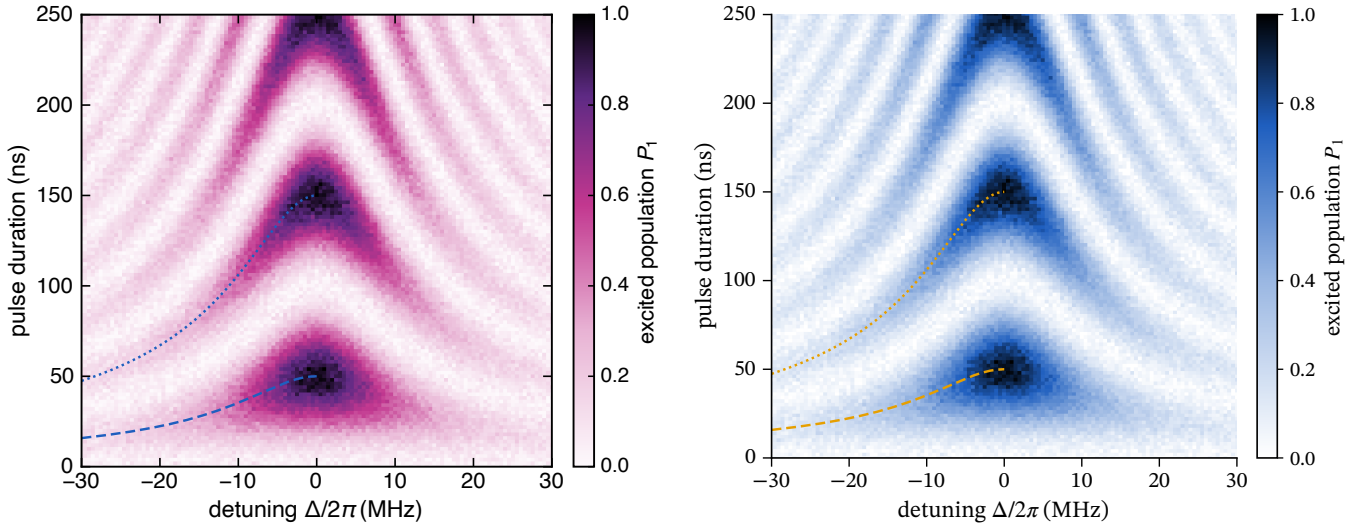
Power Rabi: excited population against drive amplitude for a 40 ns pulse, with a cosine fit and the pi-pulse amplitude (synthetic data).



32 Rabi chevron

`32_chevron.py`

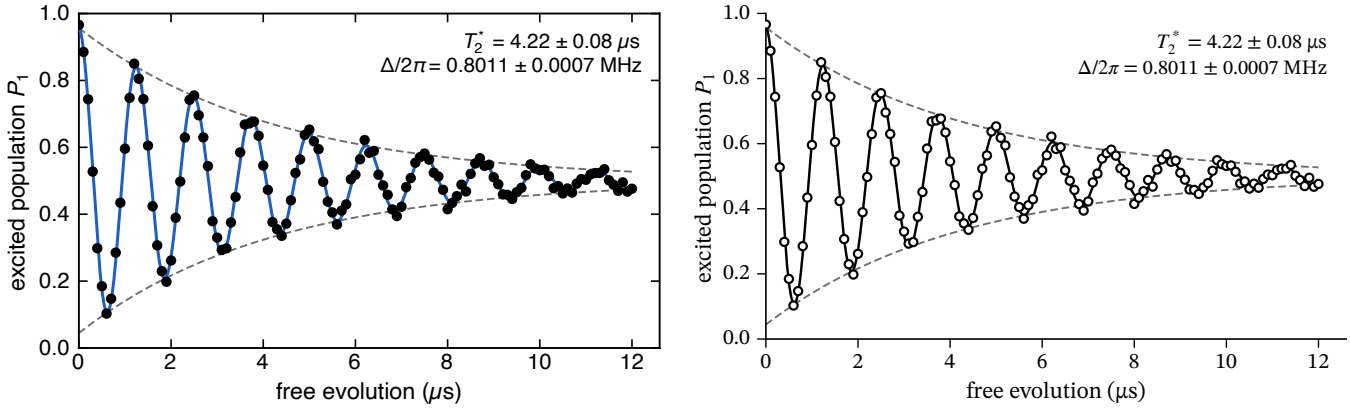
Rabi chevron: excited population against drive detuning and pulse duration at a 10 MHz Rabi rate, with the first two population maxima of the model drawn over the left half (synthetic data).



33 Ramsey fringes

`33_ramsey.py`

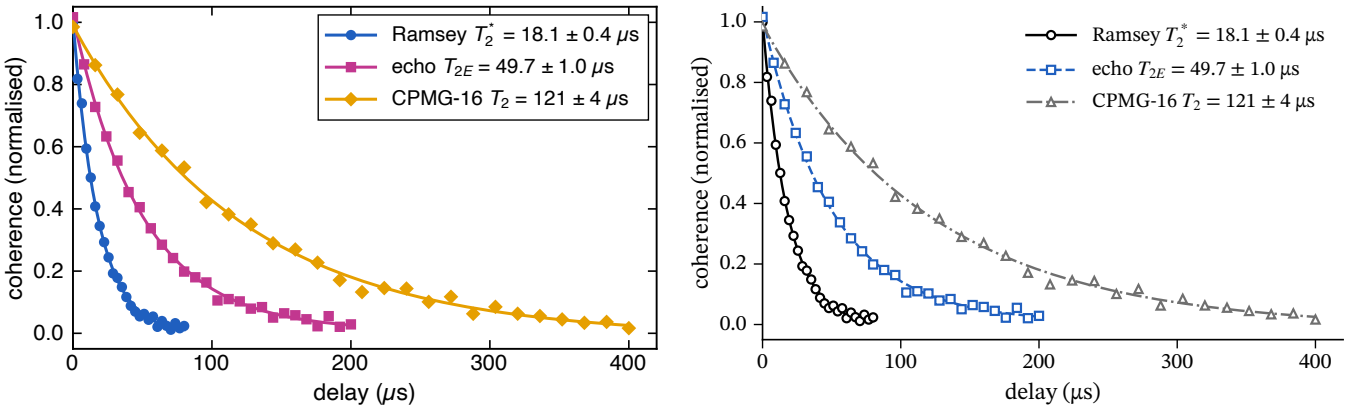
Ramsey fringes at 0.8 MHz detuning with a damped-cosine fit and its envelope (synthetic data, T2* = 4.2 us).



34 Coherence decays

`34_coherence_decays.py`

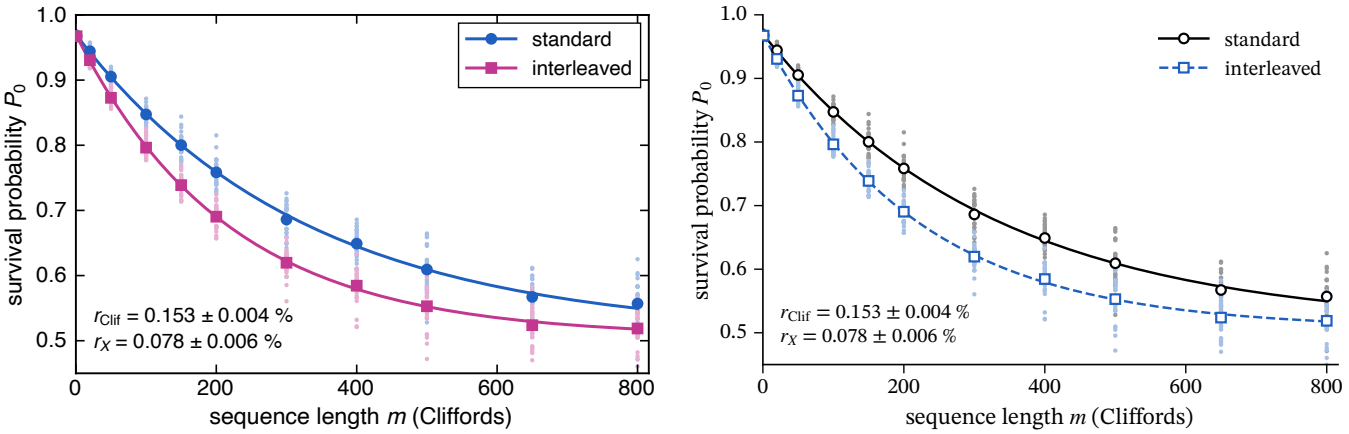
Ramsey envelope, Hahn echo and CPMG-16 decays of one qubit on one axis, each with its exponential fit and time constant (synthetic data).



35 Randomized benchmarking

`35_rb.py`

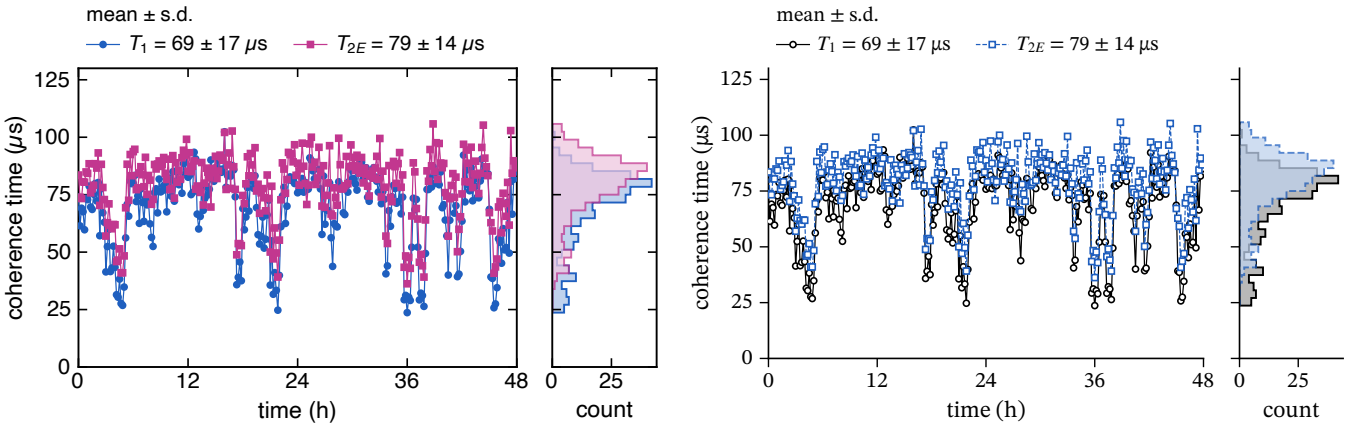
Standard and interleaved randomized benchmarking of an X gate, 30 random sequences per length, with the error per Clifford and the interleaved gate error (synthetic data).



36 Coherence over time

`36_coherence_timeseries.py`

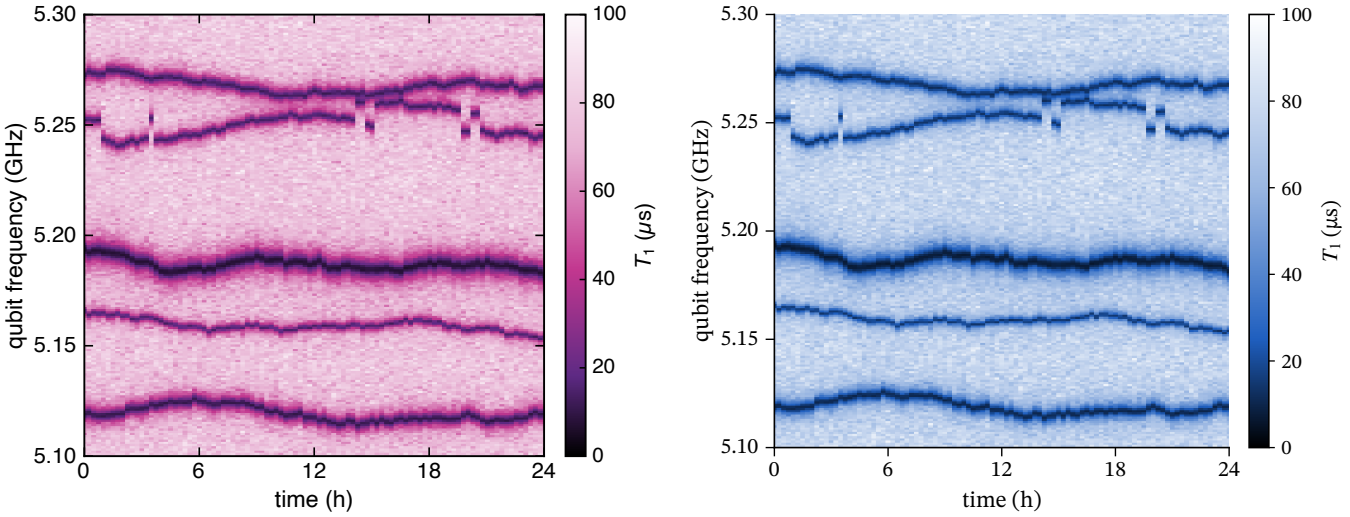
T1 and echo T2 measured every 10 minutes for 48 hours, with their histograms; T1 drops whenever a wandering defect drifts onto the qubit frequency (synthetic data).



37 TLS spectral diffusion

`37_tls_map.py`

T1 against qubit frequency over 24 hours; five defects show as Lorentzian T1 dips whose frequencies wander, and one switches between two frequencies (synthetic data).

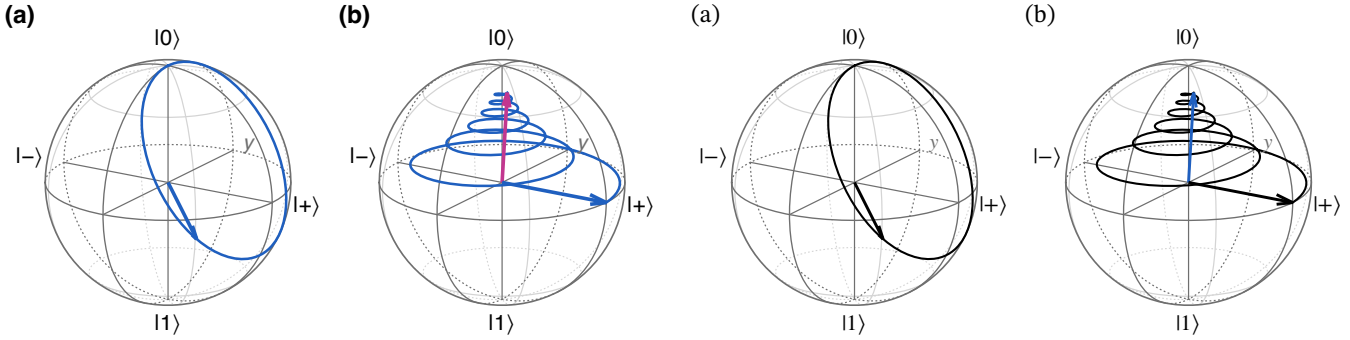


States

40 Bloch sphere

`40_bloch_sphere.py`

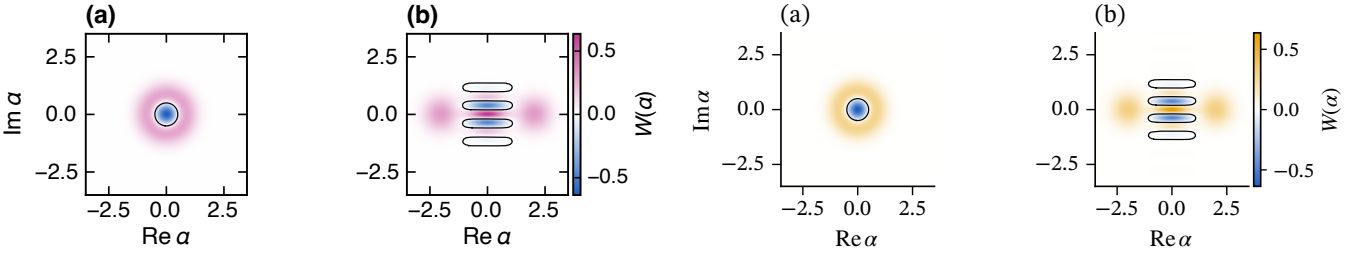
Bloch sphere: (a) a detuned Rabi drive (Delta = 0.6 Omega) tilts the precession cone off the pole; (b) free precession from |+> spirals to |0> as T2 and T1 shrink the vector (computed).



41 Wigner functions

`41_wigner.py`

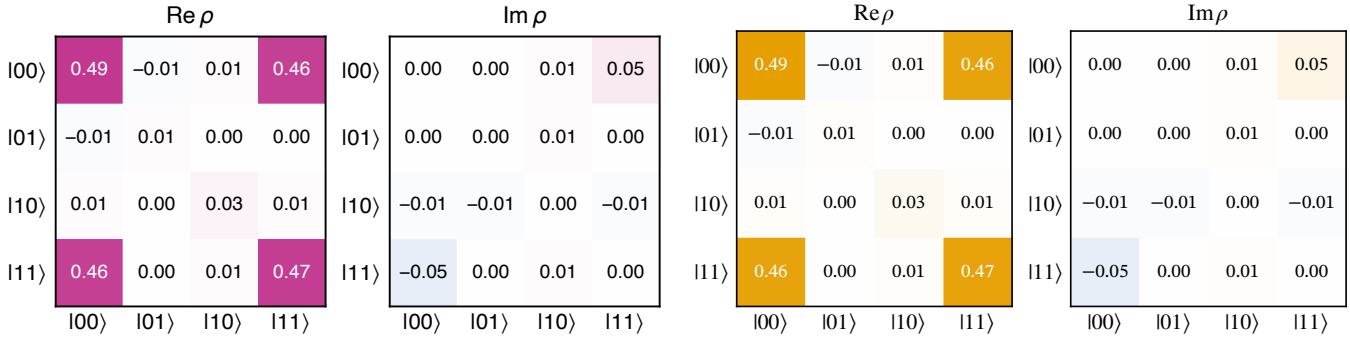
Wigner functions of (a) the Fock state |1> and (b) an even cat state with alpha = 2, on one colour scale of +-2/pi (computed).



42 Density matrix

`42_density_matrix.py`

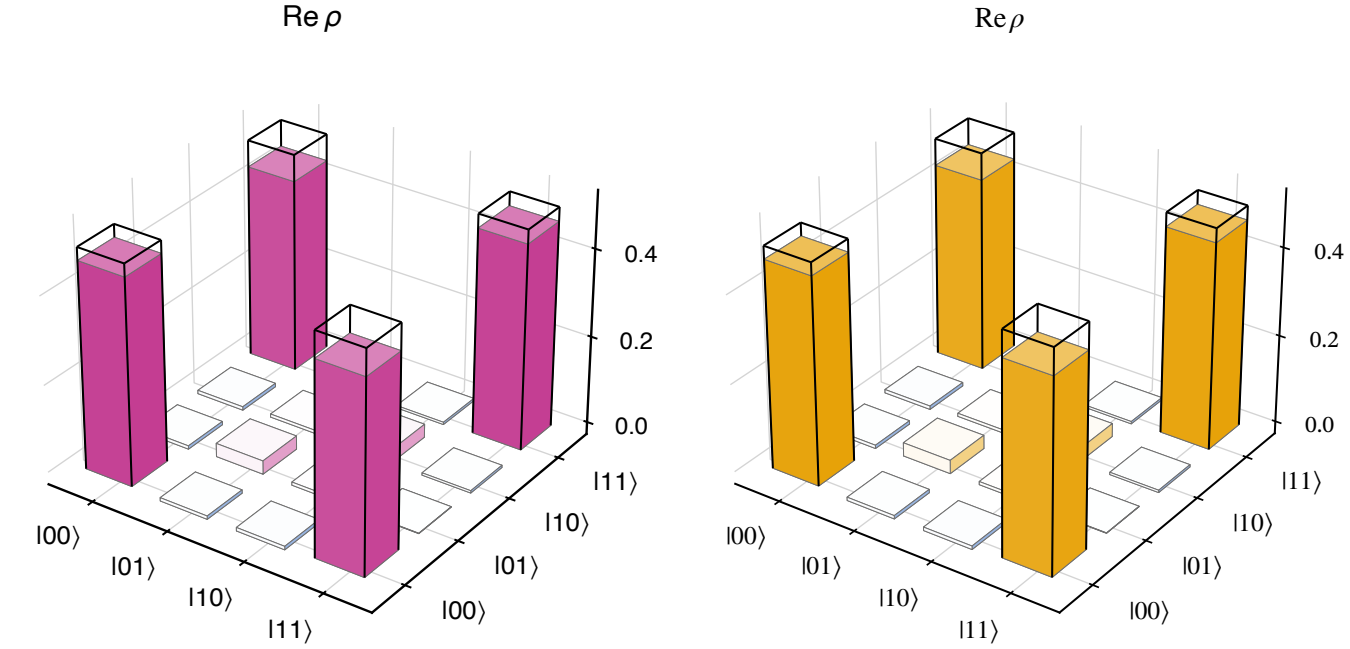
Two-qubit tomography of the Bell state (|00> + |11>)/sqrt(2): real and imaginary parts of the reconstructed density matrix (synthetic data).



43 Density matrix in 3D

`43_density_bars3d.py`

Real part of the reconstructed Bell-state density matrix as 3D bars, with the ideal state as wireframe outlines (synthetic data).

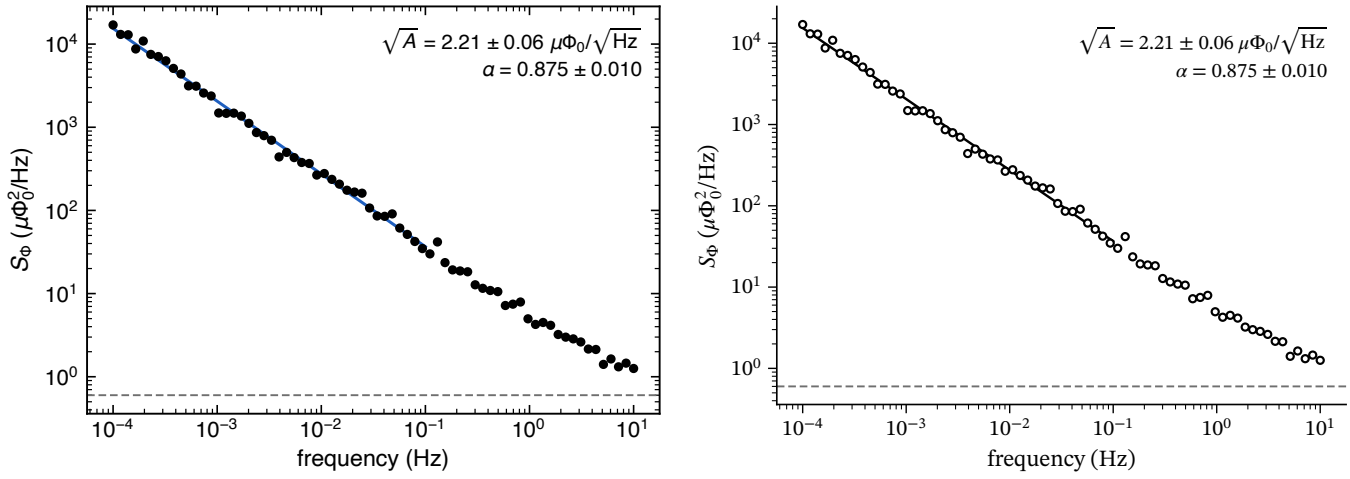


Noise and errors

50 Noise spectral density

`50_noise_psd.py`

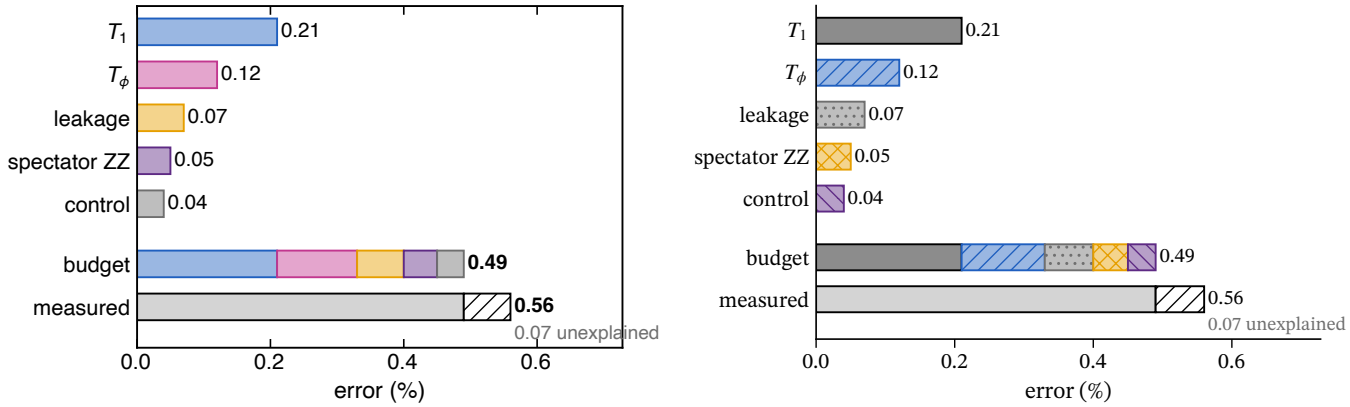
Flux-noise spectral density with a power-law fit, amplitude at 1 Hz and a white floor (synthetic data).



51 Error budget

51_error_budget.py

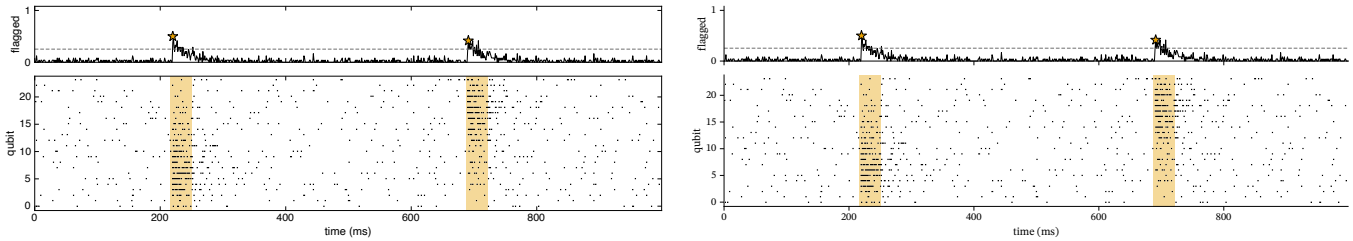
CZ-gate error budget: contributions, their sum, and the measured error with the unexplained part hatched.



52 Correlated error bursts

52_burst_raster.py

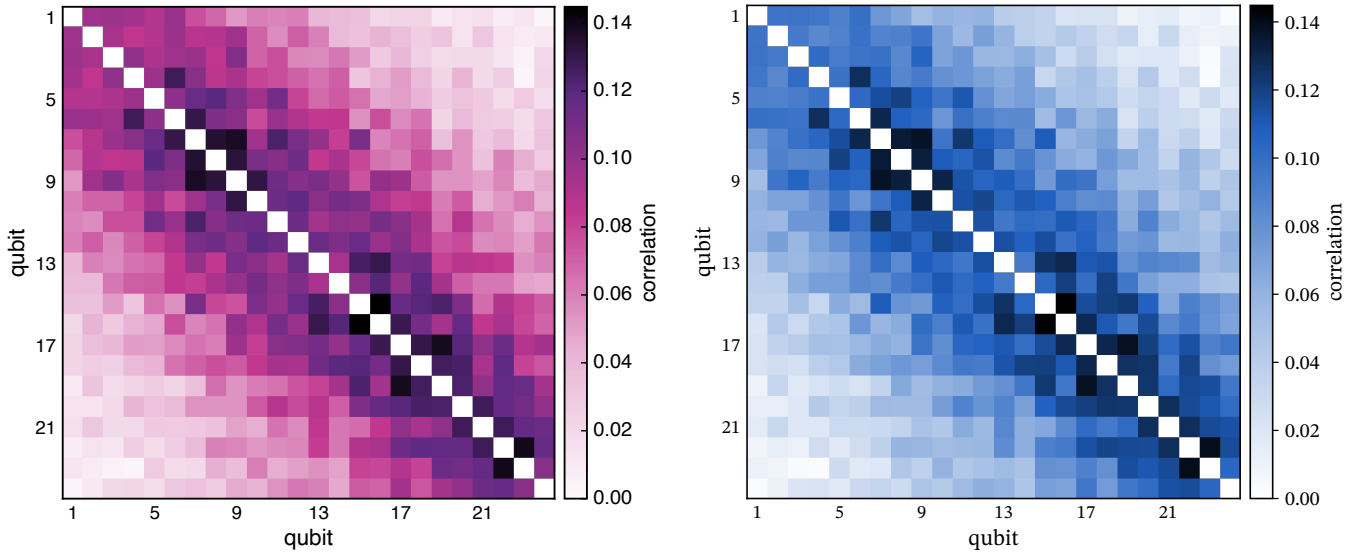
Correlated error bursts: detections per qubit over time, with the flagged fraction and burst marks above (synthetic data from a toy burst model).



53 Error correlations

53_error_correlation.py

Pairwise correlation of error detections across a 24-qubit chain; bursts correlate neighbours (synthetic data from a toy burst model).

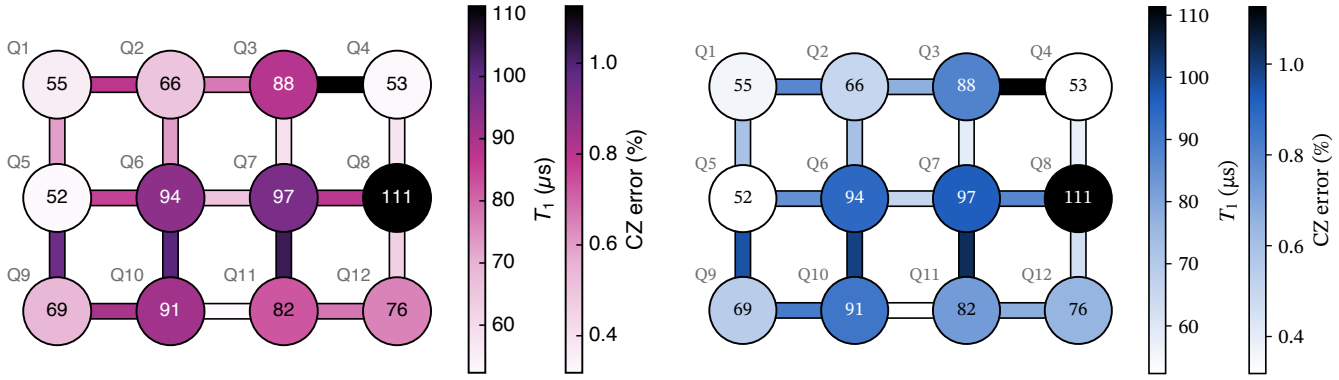


Processor

60 Device map

60_device_map.py

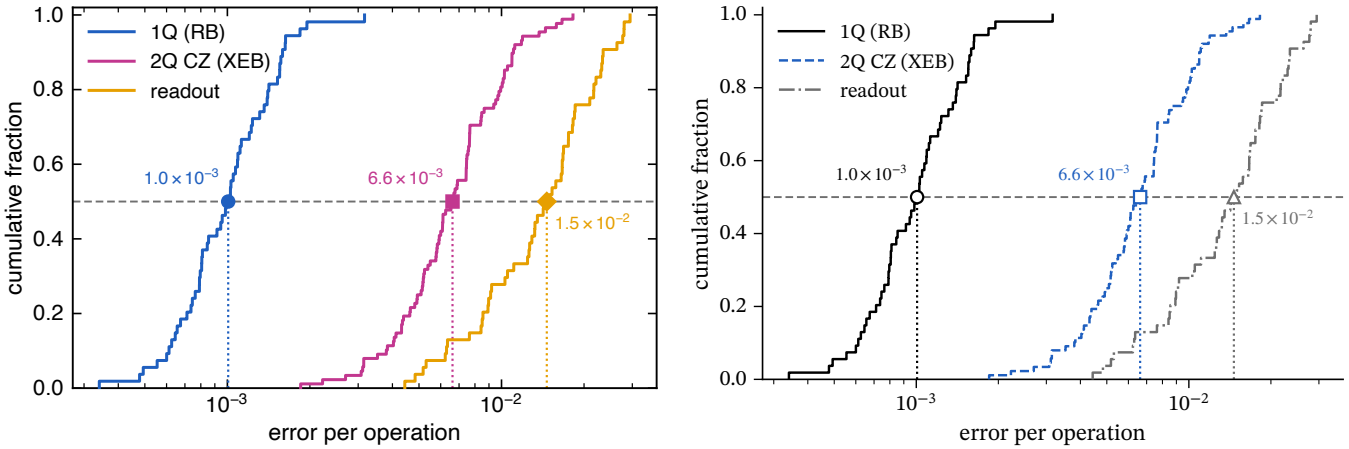
Device map of a 3 x 4 qubit grid with T1 on the qubits and CZ error on the couplers (synthetic data).



61 Error distributions

61_error_cdf.py

Cumulative distributions of single-qubit, CZ and readout errors across a 54-qubit processor, medians marked (synthetic data).

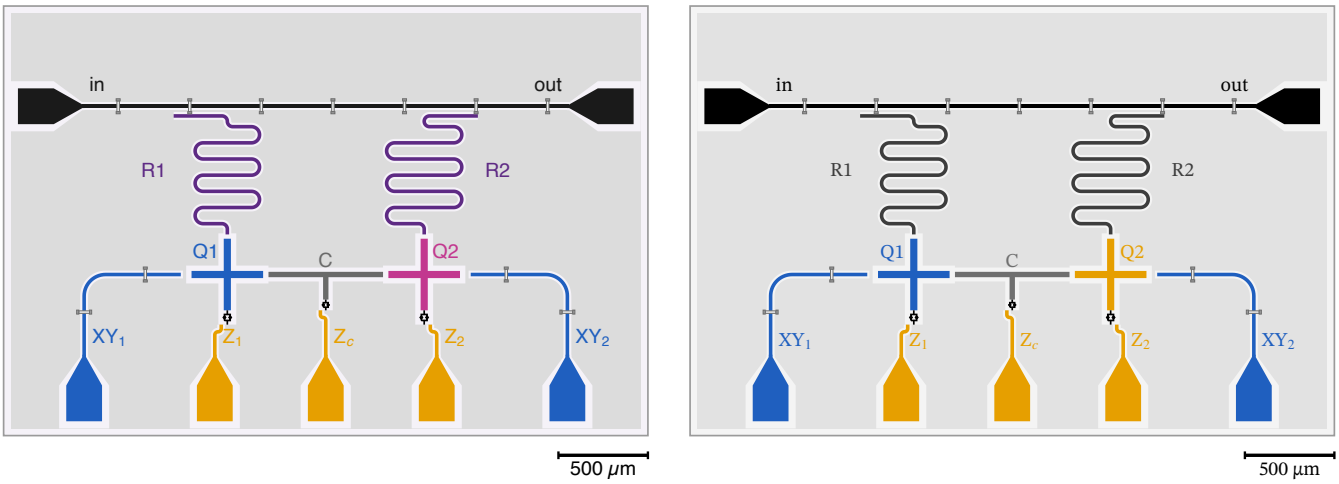


Schematics

70 Chip plan

70_chip_plan.py

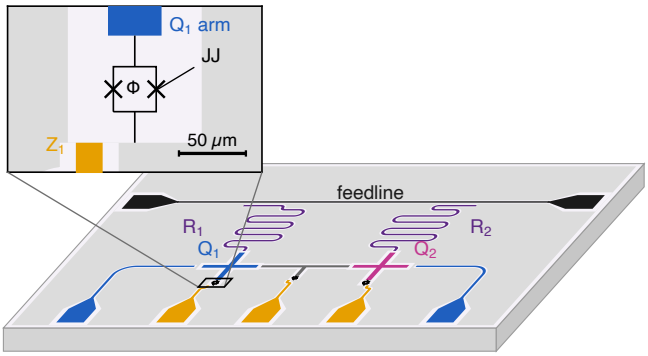
Plan view of a two-qubit chip: Xmons joined by a tunable coupler, readout resonators on a feedline, drive and flux lines, airbridges.

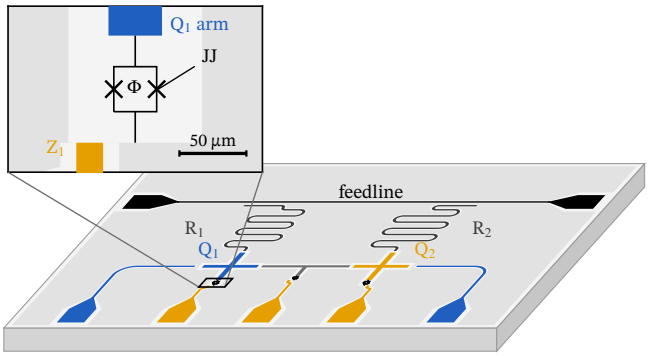


71 Chip on a slab

`71_chip_oblique.py`

The two-qubit chip of example 70 laid on a slab by one rigid oblique map, with a plan-view zoom on the SQUID of Q1.

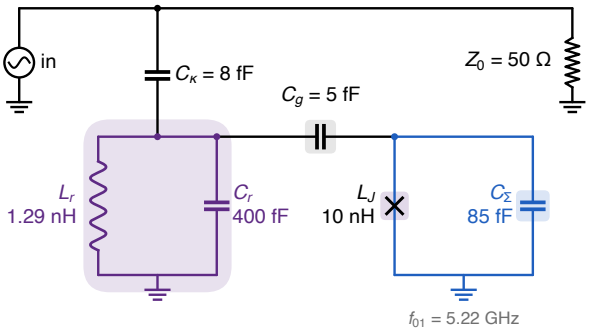




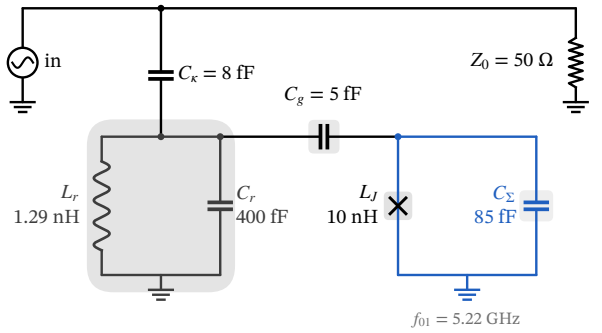
72 Transmon and readout circuit

`72_lumped_circuit.py`

Lumped circuit of a transmon coupled to a quarter-wave readout resonator on a feedline, each Hamiltonian term on the tile of the element that produces it.



$\hat{H} =$ $4E_C\hat{n}^2$ (E_C/h = 228 MHz) + $-E_J\cos\hat{\phi}$ (E_J/h = 16.3 GHz) + $hf_r\hat{a}^\dagger\hat{a}$ (f_r = 7.00 GHz) + $hg\hat{n}(\hat{a}+\hat{a}^\dagger)$ (g/h = 82 MHz)

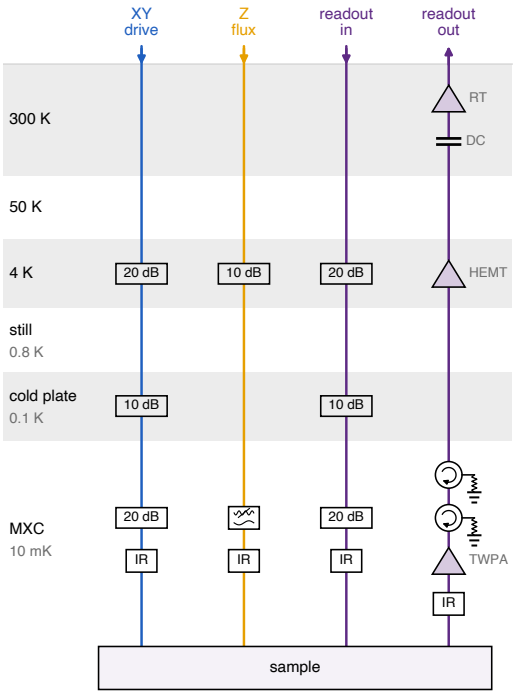


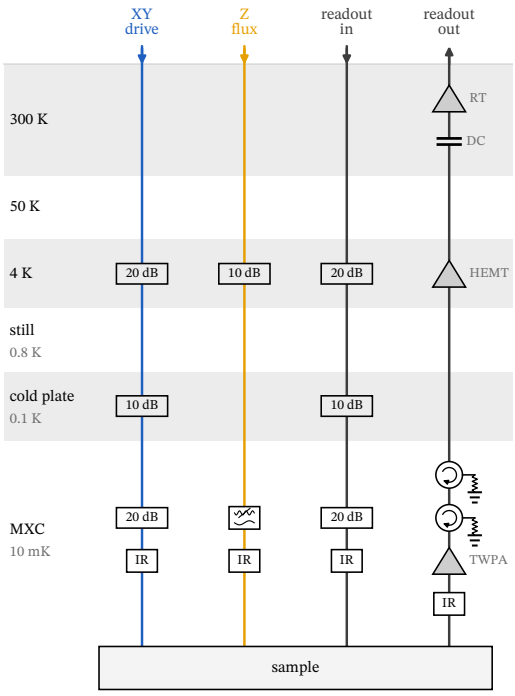
$\hat{H} =$ $4E_C\hat{n}^2$ (E_C/h = 228 MHz) + $-E_J\cos\hat{\varphi}$ (E_J/h = 16.3 GHz) + $hf_r\hat{a}^\dagger\hat{a}$ (f_r = 7.00 GHz) + $hg\hat{n}(\hat{a}+\hat{a}^\dagger)$ (g/h = 82 MHz)

73 Cryostat wiring

`73_cryostat_wiring.py`

Wiring of a dilution refrigerator: attenuated and filtered XY, Z and readout-input lines, and a readout output through isolators, a TWPA and a HEMT.

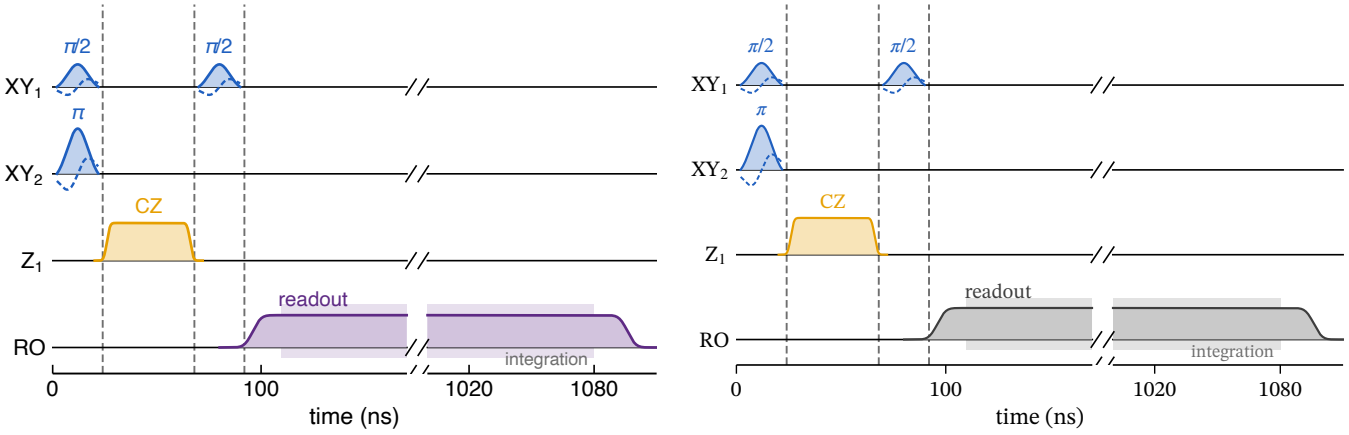




74 Pulse sequence

`74_pulse_sequence.py`

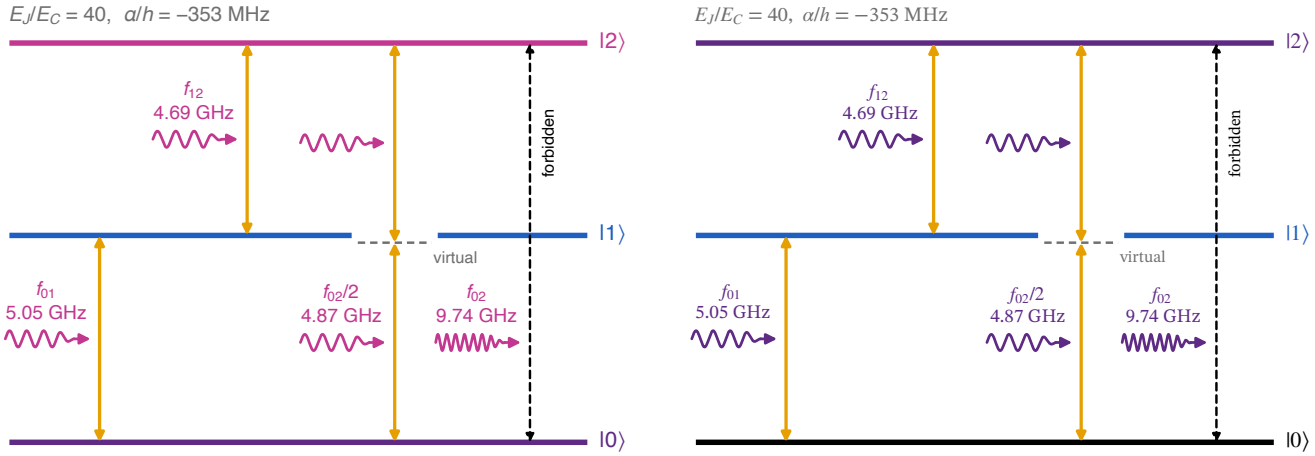
Pulse sequence of a conditional-phase measurement: DRAG pulses on two qubits, a CZ flux pulse and a dispersive readout with its integration window.



75 Level ladder and transitions

`75_level_ladder.py`

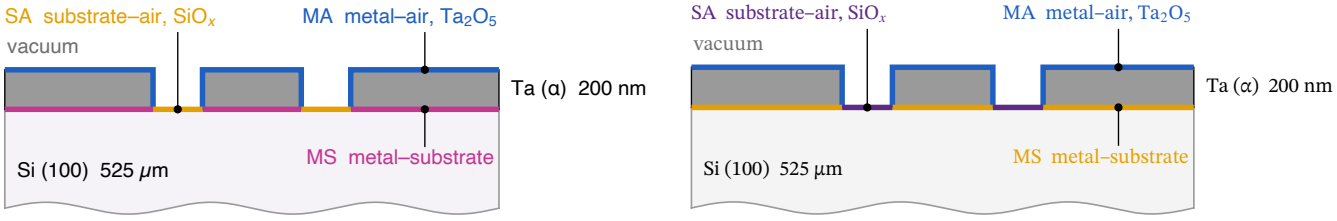
Transmon ladder (E_J/E_C = 40) to scale, with drive photons for f01, f12 and the two-photon f02/2, and the parity-forbidden single-photon 0-2 transition.



76 Film stack

`76_film_stack.py`

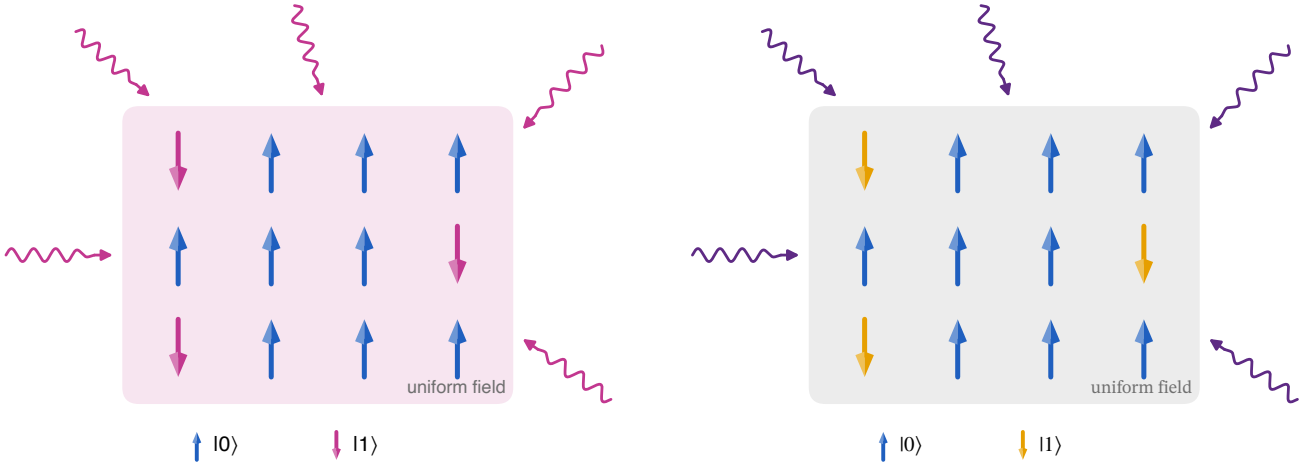
Cross-section of a tantalum CPW on silicon with its metal-air, metal-substrate and substrate-air interfaces marked.



77 Qubit array in a uniform field

`77_spin_array.py`

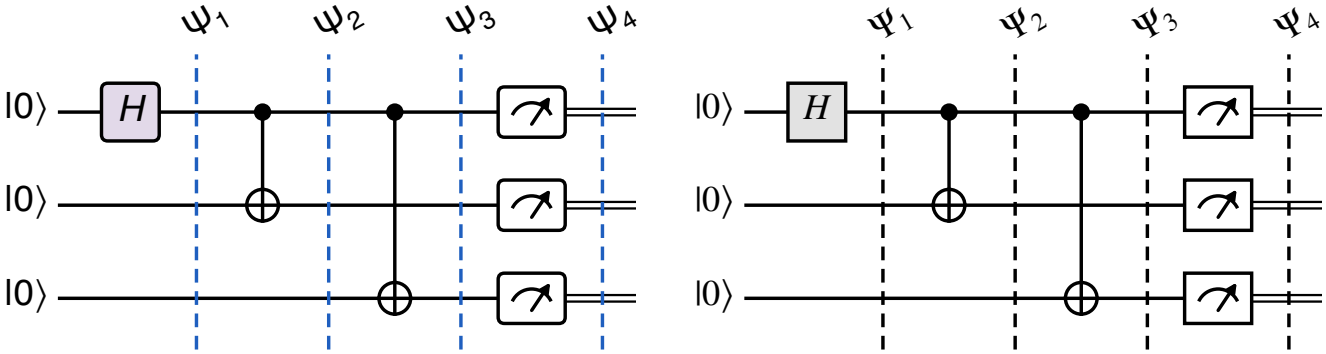
A 3 x 4 qubit array drawn as spins in a uniform field (the wash), with photons arriving from several directions and three qubits excited.



78 Quantum circuit

`78_quantum_circuit.py`

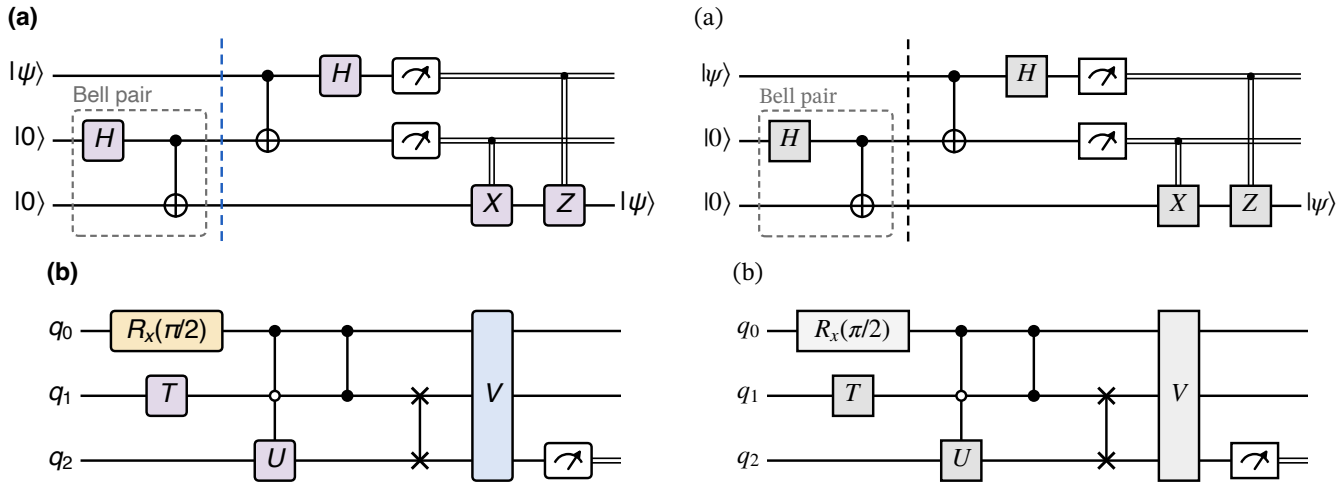
GHZ-state preparation: a Hadamard and two CNOTs entangle three qubits, then all three are measured; dashed slices mark the states Psi_1 to Psi_4.



79 Teleportation and gate glyphs

`79_teleportation.py`

(a) Teleportation: a Bell pair, a Bell measurement and classically controlled X and Z corrections. (b) The other gate glyphs: rotation, controlled-U with an open control, CZ, SWAP and a three-qubit box.



80 Black-box circuit

80_black_box_circuit.py

Black-box quantization in four steps for the device of example 72: (a) the transmon coupled to its readout resonator; (b) the junction split into its linear inductance and its nonlinear remainder; (c) the linear part as one admittance, probed by a swept current source; (d) that admittance beside the nonlinear element, whose zero-point phase each mode sets (computed).

